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IOT based smart cold chain temperatur monitoring and alert system for vaccination container

Abstract, A vaccination is one of primary precautionary source to prevent the severe effects of corona virus. The vaccination has to be maintained in the temperature range of 2°C to 8°C. The proposed system measures and monitors the temperature inside the vaccine container. A thermocouple device and smart controller encapsulated inside the container. The thermocouple senses the temperature in container. The smart WiFi enabled controller measure the temperature using electrical output of thermocouple. The measured temperature is passed to the cloud server and mobile view. The data and time wise maintenance of vaccine temperature, emergency indication, and current vaccine container temperature are displayed in the cloud. The system is useful on maintaining vaccine at prescribed temperature level and effectively protects the human life. The system achieved the accuracy of 99.5%.

Streszczenie. Szczepienie jest jednym z podstawowych środków ostrożności, aby zapobiec poważnym skutkom koronawirusa. Szczepienie należy prowadzić w zakresie temperatur od 2°C do 8°C. Proponowany system mierzy i monitoruje temperaturę wewnątrz pojemnika na szczepionkę. Termopara i inteligentny sterownik zamknięte w pojemniku. Termopara wyczuwa temperaturę w pojemniku. Inteligentny sterownik obsługujący WiFi mierzy temperaturę za pomocą wyjścia elektrycznego termopary. Zmierzona temperatura jest przekazywana do serwera w chmurze i widoku mobilnego. Dane i czasowe utrzymywanie temperatury szczepionki, wskazanie sytuacji awaryjnej i aktualna temperatura pojemnika na szczepionki są wyświetlane w chmurze. System jest przydatny w utrzymywaniu szczepionki w przepisanej poziomie temperatury i skutecznie chroni ludzkie życie. System osiągnął dokładność 99,5%. *(Inteligentny system monitorowania temperatury w łańcuchu chłodniczym oparty na IOT i system ostrzegania o pojemniku na szczepienia)*

Keywords: COVID, Vaccine, cold chain, NodeMCU, WiFi

Słowa kluczowe: szczepienie, monitorowanie temperatury

Introduction

The COVID'19 (corona virus disease) changes the life style of every human. There are certain protection systems followed to prevent the infection of COVID pneumonia. The external protection systems such as face mask, Personal Protective Equipment (PPE) cloth, hand gloves, eye glass, and sanitizers. The internal protection system developed using vaccine. The vaccination will help to protect the people from severe lung infection. The vaccinations are to be maintained with the temperature of 2°C to 8°C as per the guidelines of World Health Organization (WHO). The effect of lower temperature maintenance on vaccination leads to in effective. The ice cube container is presently used to maintain the temperature. In urban health centers are having freezers. The rural health centers need the transportation of vaccine for longer distance. The transport vehicle is equipped with freezer container. The vaccine storage locations are also having the vaccine freezing facility.

Literature review

Yakum et al. (2015) [1] have analyzed the ineffectiveness in the vaccines due to lack of temperature maintenance. The vaccine container temperature is measured physically at various locations of North western region, Cameroon. The major reason identified for lack in temperature measurement is due to power failures. An importance of supply chain management, temperature maintenance of various vaccines and methodologies to be adopted were discussed by Bajrovic et al. (2020) [2]. Hanson et al. (2017) [3] have given review on the effect of ineffectiveness of vaccine due to loss of temperature. The vaccines are to be maintained with improper temperature conditions leads to makes the severe effects on children. The broad survey was conducted which concluded that the major reason for vaccine improper maintenance is due to lack of concentration of workers. The vaccine cold chain leads to loosing many countries investment. Maglasang et al. (2018) [4] have discussed the requirement of freezer and refrigerator requirement of rural health centers. The

requirements of vaccine fertilization at the time of transportation were analyzed. The monitoring of vaccine temperature and power failures were monitored mechanically. The study on nature of vaccine effects get changed due to abnormal temperature was analyzed by Sow et al. (2018) [5]. The refrigeration failure leads to reduces the effectiveness of vaccines and also the color of vaccine also gets changed. Yamoah et al. (2019) [6] have proposed the detailed study on vaccine cold chain maintenance. The effect of vaccine performance at different age group was proposed by Ateudjieu et al. (2020) [7]. The children from the 0 to 59 months age group children vaccination effectiveness were studied. A mobile network based remote level temperature monitoring was proposed by Lutukai et al. (2019) [8]. A cold chain maintenance of vaccine temperature in urban center analogy was made by Ringo et al. (2017) [9]. An electronic controller based breathing level monitoring was proposed by thiyaneswaran et al. (2020) [10]. The causes of breaking cold chain mechanism of medicine were discussed by das (2020) [11]. The cloud based air quality updation were proposed by Thiyaneswaran et al. (2021) [12].

Proposed cold chain monitoring

There is system required to monitor the storage temperature of vaccine. The block diagram of proposed system is shown in the Fig.1. The proposed system has the measurement unit along with the Internet of Things (IoT) facility. The nodeMCU has facility of inbuilt WiFi module. The temperature is measured using the PT100 thermistor sensor device notified in the Fig.1. It has able to sense the temperature in range of -150° to +150°. A rechargeable battery backup is made with the capacity of 1000 mAH. The proposed system will come up to backup time of 3 days (72 hours). The complete setup is packed in plastic non-interference container. The steel part of sensor module is projected in cooling container which consists of vaccines. The sensor senses the temperature and converts into equivalent analog electrical signal. The analog electrical signal is converted into digital quantity by the controller. The

Table 1. Measurement of temperature at various locations

S.No	Location	Type of container	Temperature measurement using standard meter	Temperature measurement using proposed system
1	PHC-01	Container	1.5°C	1.5°C
2	PHC-02	Cold chain refrigerator	1°C	1.1°C
3	PHC-03	Cold chain refrigerator	2°C	2.1°C
4	PHC-04	Container	2.5°C	2.5°C
5	PHC-05	Container	3°C	3°C
6	PHC-06	Cold chain refrigerator	2.3°C	2.3°C
7	PHC-07	Container	4°C	4°C
8	PHC-08	Container	3.5°C	3.5°C
9	PHC-09	Cold chain refrigerator	2°C	2°C
10	PHC-10	Container	2.8°C	2.8°C

Table 2: Comparison of accuracy with existing system

S.No	System Description	Refrigeration Maintenance Percentage
1	Proposed system (2022)	99.5%
2	Bajrovic et.al (2020)	97.3%
3	RTM, Lutukai et al (2019)	97%
4	Automated Recording System, Sow et al (2018)	98%

The proposed system accuracy is compared with the existing temperature monitoring system shown in the Table-2. The proposed system accuracy is about 99.5%. The drop in accuracy is due to drop in rural area mobile signals. The accuracy of the proposed system is 1.5% to 2.5% greater than the existing methods.

Conclusion

The cold chain system is very important on maintaining the vaccine. The system is required at the time of transporting vaccines from one country to other country, one state to another state, state to districts, district to urban, and rural centers. In the transportation period vaccines temperature has to be maintained under the limit. The vaccines are made in remote area in establishment of smaller campus and under these situations also the temperature of vaccines to be maintained. At level storage points, urban and rural centers temperature is maintained using refrigerators or freezers. In case of small camp areas vaccine cooling system is maintained using ice cubes or ice bag containers. The temperature of refrigeration, freezers, smaller ice cube containers are to be monitored periodically. The proposed system consists of compact hardware concealed in the non-inference container. The container is placed inside the cooling system. A thermostat lead is dipped in container used to sense the temperature. The temperature is effectively measured and computed using the NodeMCU smart WiFi enabled controller. The temperature is updated in cloud for every 5 seconds. The alert system is useful to turn on refrigeration or used to find the failure of refrigeration system. The system will help on protecting vaccination from abnormal temperature and help the society to safe guard from COVID and other associated virus infection.

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REFERENCES

- [1] Yakum, M.N., Ateudjieu, J., Pélagie, F.R. et al. Factors associated with the exposure of vaccines to adverse temperature conditions: the case of North West region, Cameroon. BMC Res Notes 8, 277 (2015).
- [2] Bajrovic, I., Schafer, S. C., Romanovicz, D. K., & Croyle, M. A. (2020). Novel technology for storage and distribution of live vaccines and other biological medicines at ambient temperature. Science Advances, 6(10), eaau4819.
- [3] Hanson, C. M., George, A. M., Sawadogo, A., & Schreiber, B. (2017). Is freezing in the vaccine cold chain an ongoing issue? A literature review. Vaccine, 35(17), 2127–2133.
- [4] Maglasang, P. L., Butalid, M. L., Pastoril, M. F., Pratama, A. N., & Tan, E. Y. (2018). A cross-sectional survey on cold chain management of vaccines in Cebu, Philippines. Pharmacy Practice, 16(2), 1167.
- [5] Sow, C., Sanou, C., Medah, C., Schlumberger, M., Mireux, F., Ouédraogo, I., Ouédraogo, S., & Betsem, E. (2018). Challenges of cold chain quality for routine EPI in south-west Burkina-Faso: An assessment using automated temperature recording devices. Vaccine, 36(26), 3747–3755.
- [6] Yamoah, Peter, Varsha Bangalee, and Frasia Oosthuizen. 2019. "Knowledge and Perceptions of Adverse Events Following Immunization among Healthcare Professionals in Africa: A Case Study from Ghana" Vaccines 7, no. 1: 28.
- [7] Ateudjieu, J., Yakum, M., Gaura, A.P. et al. EPI immunization coverage, timeliness and dropout rate among children in a West Cameroon health district: a cross sectional study. BMC Public Health 20, 228 (2020).
- [8] Lutukai, M., Bunde, E. A., Hatch, B., Mohamed, Z., Yavari, S., Some, E., Chweya, A., Kania, C., Ross, J. C., Keddem, C., & Chandani, Y. (2019). Using Data to Keep Vaccines Cold in Kenya: Remote Temperature Monitoring With Data Review Teams for Vaccine Management. Global Health: Science and Practice, 7(4), 585–597.
- [9] Ringo, S., Mugoyela, V., Kaale, E., & Sempombe, J. (2017). Cold Chain Medicines Storage Temperature Conformity by the World Health Organisation in Tanzania. Pharmacology & Pharmacy, 08(10), 325–338.
- [10] Thiyaneswaran, B., Bhuvaneshwaran, V., Dharun, M.S., Gopu, K., Gowikan, T. (2020). Breathing Level Monitoring and Alerting by Using Embedded IOT, Journal of Green Engineering, 10(6), 2986 –2994.
- [11] Das, M. K., Arora, N. K., Mathew, T., Vyas, B., Devi, S. K., & Yadav, A. (2020). Temperature integrity and exposure of vaccines to suboptimal temperatures in cold chain devices at different levels in three states of India. Tropical Diseases, Travel Medicine and Vaccines, 6(1).
- [12] B. Thiyaneswaran, P. Elayaraja, P. Srinivasan, S. Kumarganesh, K. Anguraj, IoT based air quality measurement and alert system for steel, material and copper processing industries, Materials Today: Proceedings, 2021, ISSN 2214-7853